Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/09/2007
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Project Title:	Optimizing Science From a STEP Mission		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2911		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		
Task Description:	Study STEP signal fine structure due to oblateness of earth and earth surface irregularities; and relate this Equivalence Principle violating fine structure signal to theory. Add additional scientific information about fundamental physical law which can be achievable from a STEP mission. This mission tests one of the fundamental properties of physical law, and additional information from fine structure will aid theorists in interpreting any breakdown of the Equivalence Principle.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	The original way of thinking about the STEP experiment was to consider the Earth as a spherical, monopolar source of some new interaction field which would interact with different materials in a low-orbiting spacecraft in a manner which produced different accelerations in the various substances. Of course, this is still the correct way to view the dominant possibility of any Equivalence Principle violating (EPV) interaction due to the Earth's matter, but the Earth at the part in a thousand level is non-spherical and composed of different materials, itself. Since STEP is planned to test the universality of gravitational free fall by five orders of magnitude higher precision than present tests, the question of this research project has been what other information could potentially be obtained from the STEP experiment because of the Earth's deviations from a monopolar, compositionally homogeneous source? The basic property of an EPV interaction is that matter responds to its force field with a proportion which differs from its mass density. But from the principle of action equals reaction in field theories of interaction, it follows that for such an EPV interaction matter elements will be a source for the interaction field with strengths differing from their mass densities as well. This property of any EPV interaction between Earth and STEP satellite is the foundation for what this study explored. Because of the Earth's rotation its matter distribution is globally distorted into a non-spherical shape which then produces a significant Newtonian quadrupole gravitational field. This field's strength is parameterized with a coefficient J2 which is about 1/1000 and is related to the degree of flattening of the Earth and its matter's average squared distance from the center. A similar quadrupole EPV field should therefore exist as well. But Earth's composition consists of an iron-rich core making up about 1/3 the Earth's mass, and a surrounding mantle and crust composed of elements much lower in the periodic table. In the production of the EPV quadrupole moment of the Earth, the mantle and crustal material will contribute in greater proportion to its mass contribution than will the iron core. The result is that the EPV quadrupole moment coefficient K2 will in general differ from the well known J2 coefficient of Earth, and in proportion to the difference in coupling strength per unit mass of iron and the light elements in producing the EPV interaction. A quadrupole EPV field of Earth will produce differential accelerations on the STEP proof masses with different frequencies than will the monopolar EPV field. So the mission can separately search for these two signals and potentially measure an independent property of any EPV interaction that it may detect. We then turned to consideration of the Earth's crust, by itself, which represents a couple parts in a thousand of the Earth's total mass. This then layer of matter strongly breaks the spherical symmetry of an idealized Earth source and on a scale of relatively short distances. The composition of the top 10 or 20 kilometers of the Earth varies strongly as we range from deep ocean basins to continental shelves to high plateau and mountain regions. And because a STEP spacecraft will orbit close to the Earth's surface, it will respond to any EPV signals originating especially from the restricted segment of the planet's surface region directly below the spacecraft moment by moment. Any EPV signal due to the Earth's crust material will vary so rapidly and with the irregular nature of the terrain along the spacecraft track, a multipolar representation of this field is not useful, but rather directly expressing such a field in terms of a geological/geographical model of the Earth's surface is used. One might ask whether these crust-dependent EPV signals can be large enough to measure in a STEP mission. Beyond the Newtonian quadrupole moment of the Earth it is known that the Newtonian multipolar moments are of order a part in a million or smaller. But the smallness of the Newtonian moments is in large part due to isostacy in which the various segments of the Earth's crustal materials are close to floating on the underlying mantle material, displacing an amount of the latter equal to their own masses. This to lowest approximation eliminates irregularities of the Earth's crust from contributing to the multipole moments of the planet's Newtonian gravity. But if the displaced mantle materials produce an EPV field with source strength in different proportion to their mass densities than the crustal materials produce an EPV field not neutralized by isostacy, resulting in much larger multipolar field contributions to the EPV field which the STEP spacecraft can measure. In order for a STEP mission to measure this fine structure of an EPV signal, the design and instrumentation will need to have the flexibility to respond to a broad band of higher signal frequencies in addition to the main orbital period frequency. Publications which grew out of this project. The third is on another aspect of physical law, but was stimulated by exploring the consequences of angular asymmetries in the matter composition of the Earth. STEP's equivalence principle violating force field due to non-spherical Earth. Classical and Quantum Gravity 17 (2000) 5531-2536 Earth's equivalence principle violating multipoles: more science from a robust violation signal in STEP. Classical and Quantum Gravity 18 (2001) 2467-2473 Testing Newton's third law using lunar laser ranging. Classical and Quantum Gravity 18 (2001) L133-L137
Bibliography Type:	Description: (Last Updated: 12/14/2020)
Articles in Peer-reviewed Journals	Nordtvedt K. "STEP's equivalence principle violating force field due to non-spherical Earth." Classical and Quantum Gravity 2000;17(13):2531-6. , Jan-2003
Articles in Peer-reviewed Journals	Nordtvedt K. "Earth's equivalence principle violating multipoles: more science from a robust violation signal in STEP." Classical and Quantum Gravity 2001;18(13):2467-73. , Jan-2003
Articles in Peer-reviewed Journals	Nordtvedt K. "Testing Newton's third law using lunar laser ranging." Classical and Quantum Gravity, 2001;18(20):L133-L137. , Jan-2003